

Repair Approval Memo —

RAM-2025-004

CASCADIA RENEWABLE OPERATIONS LLC

Internal Memorandum — Operations, Engineering, and Procurement

Field	Value
Document ID	RAM-2025-004
Date	2025-12-29
From	Marcus Chen, Sr. Reliability Engineer · co-signed: Priya Ramachandran, Plant Manager, NW-PowerPool
To	Elena Marquez, VP Engineering · Daniel Voss, Director of Operations · cc: Lillian Park, CFO Office · Jeremy Holcomb, Vendor Management Lead
Subject	ESCALATION — Apex Emergency Invoice EMR-2025-006 (TRA-10561, \$402K); request engineering working group on transformer replacement-vs-refurbish
EPA Reference	EPA-2024-11 (invoked 2025-12-10)
Distribution	Engineering Steering · Operations Leadership · Vendor Management · CFO Office

1. Background

On 2025-12-10, TRA-10561 (Siemens Energy 75 MVA power transformer, NW-PowerPool-WIND-003) tripped on overheating (outage OUT-2025-1407). The forced outage ran 215 hours and produced **3,558.6 MWh of lost generation**. CPS could not mobilize within the EPA-2024-11 window (winter weather across the Cascades); Apex Emergency Grid Solutions was dispatched and Siemens Energy OEM engineering attended on-site.

This is the second emergency repair on TRA-10561 in 6 months. The prior event (2025-05-26, OUT-2025-1405, vibration_excess, 158 hours, 2,615.1 MWh lost) was documented in Apex report **EMR-2025-002**, which performed mechanical wedging and an oil change to permit derated operation. That stabilization did not hold.

Apex's current report **EMR-2025-006** documents accelerated DGA progression (acetylene 14 → 41 ppm; ethylene 92 → 178 ppm; hydrogen 280 → 410 ppm), advanced partial-discharge activity, and a tank-top temperature excursion to 118 °C. **Apex declined to re-energize the unit** and recommends full transformer replacement at \$2.85M (Siemens Energy quote SE-Q-2025-414). The Apex emergency invoice for de-energization scope is **\$402,000**.

2. Vendor Recommendation

Apex and Siemens Energy on-site engineering jointly recommend:

1. Pay the \$402K emergency stabilization / de-energization invoice (EMR-2025-006).
2. Authorize **full transformer replacement** at \$2.85M, expedited procurement.
3. Do **not** re-energize TRA-10561 in any configuration pending the replacement.

Apex cites its own prior emergency work (EMR-2025-002) as "evidence of imminent failure." Cascadia engineering acknowledges that argument has technical merit on the dissolved-gas and partial-discharge trends, while also noting that prior Apex recommendations across FY2025 have consistently favored replacement.

3. Cascadia Position

Operations and Reliability Engineering recommend:

1. **Approve** the \$402K emergency stabilization / de-energization invoice. The de-energization decision was technically correct given the DGA and acoustic-emission progression. Authorization rests under EPA-2024-11.
2. **Do not, at this memo, approve the \$2.85M replacement.** TRA-10561 is in safe out-of-service condition and there is no operational urgency that overrides a controlled engineering decision. NW-PowerPool-WIND-003 generation is currently routed through the adjacent transformer with acceptable derating.
3. **Convene a transformer replacement-vs-refurbish working group** under Engineering, with mandate to:
 4. Independently evaluate the TRA-10561 oil DGA trend, partial-discharge data, and winding condition assessment.
 5. Procure a non-OEM refurbishment quote (windings rewind, core re-clamp) from a Siemens Energy-certified refurbisher.
 6. Reconcile **CSM-7** oil-spectrometer cadence with Siemens Energy Manual SE-TR-2300 §7.4 (the CSM-7 quarterly cadence conflicts with the OEM annual cadence that Apex relied on in EMR-2025-002).
 7. Report a recommended path forward (replace, refurbish, or run-to-failure with capacity firming) within 45 days.

8. **Cross-reference and elevate** the case to CAR-2025-002 (PM cadence inadequate), MOC-2025-022 (the broader OEM-vs-internal-standard reconciliation), and RCA-2025-002 (Fall Overheating Sequence root cause). The TRA-10561 case is the cleanest example in the corpus of OEM-pushed replacement against operations-preferred refurbishment.

4. Financial Justification

Line	Amount
Apex emergency invoice — EMR-2025-006 (de-energization)	\$402,000
Apex prior invoice on same asset — EMR-2025-002 (stabilization, ineffective)	\$389,000
Cumulative emergency spend on TRA-10561 FY2025	\$791,000
Siemens Energy full replacement quote SE-Q-2025-414	\$2,850,000
Refurbishment estimate (preliminary, to be confirmed)	\$950K – \$1.4M
Asset lost-generation revenue impact (TRA-10561 FY25, both events)	~\$370K at NW-PowerPool realized pricing

The cumulative emergency spend on this single transformer (\$791K) without restoring durable service is itself the principal finding of this memo. The working group's brief will include the explicit question of whether the cost trajectory justifies the OEM-recommended replacement, the refurbishment alternative, or run-to-failure with substitution capacity.

NW-PowerPool FY25 maintenance budget impact: approving the \$402K stabilization absorbs ~3.0% of the \$13.4M annual line. **Total Apex invoicing at NW-PowerPool in FY25 will close above \$4.3M** after this approval; CPS equivalent-scope estimates suggest a counterfactual cost of approximately \$1.0M (refer benchmark filed in RAM-2025-003 / CAR-2025-004).

5. Approval Request / Decision

Approved as escalated.

1. The \$402K Apex invoice EMR-2025-006 is approved for payment under EPA-2024-11.
2. The \$2.85M Siemens Energy replacement scope (SE-Q-2025-414) is **not** approved here. The decision is escalated to the new working group.

3. Engineering will charter the **TRA-10561 / Transformer Replacement-vs-Refurbish Working Group** within 14 days, with a 45-day decision deadline. Marcus Chen named technical lead; Elena Marquez sponsor.
4. CSM-7 reconciliation with Siemens Energy Manual SE-TR-2300 §7.4 is added to the working group brief and to MOC-2025-022.
5. TRA-10561 to remain de-energized; NW-PowerPool-WIND-003 to operate on the adjacent transformer at derated capacity. Operations to monitor.

Approver	Role	Decision	Date
Marcus Chen	Sr. Reliability Engineer (author)	Approve invoice; charter working group	2025-12-29
Priya Ramachandran	Plant Manager, NW-PowerPool	Concur	2025-12-29
Elena Marquez	VP Engineering	Concur — sponsor working group	2025-12-30
Daniel Voss	Director of Operations	Approve as escalated	2025-12-30
Lillian Park	CFO Office	Approve; flag \$2.85M conditional commitment in FY26 capital plan; note cumulative TRA-10561 spend	2025-12-30

References: EMR-2025-006 · EMR-2025-002 · EMR-2025-001 · EMR-2025-004 · RAM-2025-001 · RAM-2025-002 · RAM-2025-003 · CAR-2025-002 · CAR-2025-004 · MOC-2025-022 · RCA-2025-002 · Siemens Energy Manual SE-TR-2300 §7.4 · CSM-7 · MS-2025-03 · EPA-2024-11.